

Quantum Simulation of Photo-Induced Charge Dynamics: From Ethylene to Photosynthesis

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Abstract

We implement a VQE+QSE pipeline using the quantum equation-of-motion (qEOM) method as the numerically robust realization of QSE to compute the ground and low-lying excited states of ethylene (C_2H_4) in a $(2e, 2o)$ active space at the STO-3G level, benchmark it against classical CASCI, and evaluate hardware noise and error mitigation on both simulator and real IBM Quantum hardware. VQE agrees with CASCI to within 0.02 mHa; the triplet and singlet ($\pi \rightarrow \pi^*$) excited states are correctly identified; and zero-noise extrapolation with readout mitigation reduces the real-hardware ground-state error by a factor of ~ 2.8 . We close with a proposed extension toward electron-transfer rates in a chlorophyll-inspired system.

1 Introduction

Photo-induced electronic excitations underlie vision, photosynthesis, and photovoltaic energy conversion, and are classically expensive to simulate as molecules grow, motivating near-term quantum approaches such as VQE combined with QSE, realized here via the qEOM method. We use ethylene’s well-characterized $\pi \rightarrow \pi^*$ excitation as a testbed for a pipeline extensible toward larger, biologically relevant chromophores.

2 Methodology

2.1 Molecular Setup and Active Space

Ethylene was constructed at its equilibrium planar geometry (C–C = 1.34 Å, C–H = 1.087 Å, H–C–H = 117.3°, verified numerically against literature values) and at a nominally twisted geometry intended to represent 90° rotation about the C–C axis. The active space was restricted to 2 electrons in 2 spatial orbitals (the π/π^* pair) using `ActiveSpaceTransformer`, mapped to 4 qubits via the Jordan–Wigner transformation.

2.2 Ground-State VQE

Two ansätze were implemented: UCCSD (3 parameters) and a hardware-efficient `EfficientSU2` ansatz (24 parameters, linear entanglement, 2 repetitions), both optimized with COBYLA against a noiseless statevector estimator.

2.3 From QSE to QEOM

The challenge statement specifies quantum subspace expansion (QSE): diagonalize H in the subspace spanned by excitation operators applied to the VQE ground state, $\langle \psi_0 | O_i^\dagger H O_j | \psi_0 \rangle$. We implement this via Qiskit Nature’s QEOM (quantum equation-of-motion) solver rather than a bare QSE diagonalization for two reasons. First, qEOM is the numerically robust formulation of the same subspace idea: it builds the generalized eigenvalue problem from double-commutator matrices (M, Q, V, W) , which are Hermitian by construction and better-conditioned under sampling noise than directly diagonalizing $\langle O_i^\dagger H O_j \rangle$, which can become ill-conditioned or lose Hermiticity from linear dependence and finite-shot error. Second, qEOM is the version natively implemented and tested in Qiskit Nature, avoiding a from-scratch, less-validated custom QSE implementation under hackathon time constraints.

The two methods are mathematically the same subspace-expansion idea and are used interchangeably in the near-term quantum chemistry literature; single and double excitations (“sd”) were used to build the subspace on top of the converged VQE ground state.

2.4 Classical Benchmarking

CASCI(2,2)/STO-3G calculations (PySCF) were performed at both geometries as an independent classical cross-check of the VQE and QEOM results.

2.5 Noise-Aware Benchmarking and Error Mitigation

Two noise studies were carried out: (i) the converged UCCSD ground state was evaluated on real IBM Quantum hardware (`ibm_marrakesh`) at Qiskit Runtime resilience levels 0 (raw), 1 (TREX readout mitigation), and 2 (zero-noise extrapolation + TREX + gate twirling); (ii) the QEOM excited-state calculation was repeated locally under a realistic device noise model (derived from a `FakeSherbrooke` hardware snapshot) using Qiskit Aer’s `EstimatorV2`, isolating the effect of noise on the excitation-operator measurements while holding the (noiselessly-optimized) ground state fixed.

3 Results

3.1 Ground-State Energies

Table 1 summarizes the ground-state results at the planar geometry. Both ansatze agree with each other and with the classical CASCI(2,2) reference to within 0.02 mHa, confirming correct implementation: with only 2 active electrons, UCCSD is variationally exact, and CASCI is the exact diagonalization in the same active space.

Table 1: Ground-state energy, circuit resources, and classical cross-check (planar geometry).

Method	Energy (Ha)	Qubits	Depth (parameters)
UCCSD (VQE)	-77.116628	4	112 (3)
Hardware-efficient (VQE) ¹	-77.116184	4	12 (24)
CASCI(2,2) [classical]	-77.116608	—	—

3.2 Excited-State (Vertical Excitation) Energies

Table 2 reports the QEOM excited states at the planar geometry. Energy ordering and magnitude are consistent with a spin-resolved CASCI(2,2) calculation performed as a cross-check (triplet at 4.91 eV, singlet V-state at 14.24 eV, confirmed via $\langle S^2 \rangle$ expectation values), allowing unambiguous spin assignment of the QEOM roots.

Table 2: QEOM excited states, planar ethylene.

State	Energy (Ha)	ΔE (eV)	Assignment
Excited 1	-76.941013	4.779	Triplet (T_1)
Excited 2	-76.596800	14.145	Singlet V-state ($\pi \rightarrow \pi^*$)
Excited 3	-76.407974	19.283	Doubly-excited singlet

The triplet energy (4.78 eV) agrees well with the experimental vertical triplet energy of ethylene (~ 4.6 – 4.9 eV). The singlet V-state (14.1 eV) substantially overestimates the experimental value of 7.6 eV; this is a known limitation of the minimal STO-3G basis combined with a (2,2) active space for this particular ionic transition, and is not attributable to the quantum algorithm.

¹Hardware-efficient optimization reached the COBYLA iteration cap (300/300); the reported energy should be treated as a lower-confidence result pending a rerun with a larger budget.

3.3 Classical Cross-Validation (CASCI)

Table 3 lists the CASCI(2,2) results used for cross-validation.

Table 3: Classical CASCI(2,2)/STO-3G benchmark.

Geometry	E_0 (Ha)	ΔE , first excited root (eV)
Planar (0°)	-77.116608	4.91
“Twisted” (90°)*	-74.062176	1.91

*See Limitations (Sec. 4): the twisted geometry as constructed corresponds to a distorted structure rather than an ideal 90° rotation, and these values should be read qualitatively, not as a literal rotational barrier.

3.4 Noise and Error-Mitigation Analysis

3.4.1 Real-Hardware Ground-State Study (ibm_marrakesh)

Table 4 shows the converged UCCSD ground state evaluated on real IBM Quantum hardware at increasing resilience levels.

Table 4: Ground-state energy on `ibm_marrakesh` at increasing resilience levels.

Method	Energy (Ha)	Error vs. exact (mHa)
Exact (statevector)	-77.116628	0.000
Raw (resilience level 0)	-76.985608	131.02
Readout-mitigated, TREX (level 1)	-77.004366	112.26
ZNE + TREX + twirling (level 2)	-77.070221	46.41

Error mitigation reduced the deviation from the exact energy by $\sim 2.8\times$ ($131\text{ mHa} \rightarrow 46\text{ mHa}$). The raw hardware error alone ($131\text{ mHa} \approx 3.6\text{ eV}$) exceeds the entire triplet excitation energy, underscoring that mitigation is not optional for chemically meaningful results on current devices.

3.4.2 Local Noise Study of Excited States (QEOM)

Table 5 shows the QEOM excitation energies recomputed locally under shot noise only and under a realistic device noise model (`FakeSherbrooke` snapshot), with the ground state held fixed at its noiselessly-optimized value.

Table 5: QEOM excitation energies under local noise conditions.

Condition	Triplet (eV)	Singlet V (eV)
Exact	4.779	14.145
Shot noise only	4.773	14.022
Realistic device noise	4.817	14.146

The excited-state drift here ($\leq 0.15\text{ eV}$) is much smaller than the raw hardware ground-state error ($131\text{ mHa} \approx 3.6\text{ eV}$), most likely because the local model was transpiled against an all-to-all-connected simulator and omits the SWAP-gate overhead that real device connectivity imposes. This suggests routing overhead, not just intrinsic gate/readout error, dominates the real-hardware error and merits explicit modeling in future work.

4 Limitations

- **Twisted geometry:** the rotation-axis parametrization used for the twisted structure distorts C–H bond lengths rather than performing a pure rotation about the C–C axis, so the reported “twisted” CASCI energy reflects a distorted structure, not an ideal 90° ethylene, and should be read qualitatively only.
- **Spin labeling:** QEOM roots are assigned triplet/singlet character by cross-validation against a separate $\langle S^2 \rangle$ -tagged CASCI run, not by spin-purification within QEOM itself.

- **Basis set:** STO-3G with a (2,2) active space is known to overestimate ionic $\pi \rightarrow \pi^*$ energies; a larger basis/active space (e.g. $2e/4o$) would likely narrow the 14.1 vs. 7.6 eV gap.
- **Noise-model fidelity:** the local QEOM noise study omits target-backend connectivity, understating real device error.
- **Convergence:** the hardware-efficient ansatz hit its COBYLA iteration cap without an explicit convergence flag.

5 Extension: Toward Photo-Induced Charge Transfer in Larger Chromophores

Extending this pipeline toward a photosynthesis-relevant charge-transfer estimate (e.g. a chlorophyll/porphyrin fragment) requires three steps beyond what is implemented here. **(1) Active-space reduction via embedding:** a full porphyrin macrocycle is too large for near-term treatment, so we propose selecting a minimal donor-bridge-acceptor active space (frontier orbitals of a donor and acceptor site, analogous to ethylene’s π/π^* pair) via DMET or CASSCF-based selection, with the remaining macrocycle and protein/solvent environment treated at a mean-field/QM/MM level. **(2) Electronic coupling via VQE+qEOM:** the same machinery used here can, with minor extension, obtain the coupling H_{DA} between donor-localized and acceptor-localized diabatic states, e.g. via a Hadamard-test-style circuit or a generalized qEOM subspace spanning both localized configurations. **(3) Rate theory and photocurrent:** with H_{DA} , a driving force ΔG (from energy differences at donor/acceptor geometries, as we did for planar/twisted ethylene), and a reorganization energy λ (from the same pipeline at relaxed vs. vertical geometries), Marcus theory gives the transfer rate

$$k_{ET} = \frac{2\pi}{\hbar} |H_{DA}|^2 \frac{1}{\sqrt{4\pi\lambda k_B T}} \exp\left[-\frac{(\Delta G + \lambda)^2}{4\lambda k_B T}\right],$$

and chaining k_{ET} across a multi-site pigment network via a Lindblad master equation yields a steady-state flux that, scaled by charge and quantum yield, estimates a photocurrent. **Bonus:** a toy demonstration on butadiene intermediate in size between ethylene and a porphyrin fragment is a natural, tractable next step before full-scale embedding.

6 Conclusion

We implemented and validated a complete VQE+QSE (qEOM) pipeline for ethylene, cross-checked against classical CASCI, and characterized real-hardware and simulated noise alongside standard error-mitigation techniques. The ground-state result is essentially exact; the excited-state result correctly recovers the triplet energy while exposing a well-understood basis-set limitation for the singlet V-state. Error mitigation on real IBM hardware reduced the ground-state error by $\sim 2.8\times$, and we outlined a concrete path from this pipeline toward electron-transfer rates in larger, biologically relevant chromophore systems.